RETAIL PERFORMANCE / DASHBOARD DECK

What the data is, how it flows, and what is working

Every slide is a visual from the Streamlit dashboard, captured as the audience sees it.
Profit means gross profit on standard cost (pack-adjusted), not net profit.

Source: dbt marts in retail.duckdb | App: http://localhost:8501

Agenda: one tab per audience

01	Solution map	3 visuals
02	Data & model	3 visuals
03	Row flow	2 visuals
04	Management	5 visuals
05	Finance	5 visuals
06	AP / Ops	5 visuals
07	Sales	5 visuals
08	Stores	3 visuals
09	Analysis	11 visuals
10	Roadmap & AI	2 visuals

Generated from the dashboard; re-run scripts/export_dashboard_deck.py after any change.

One tile per stakeholder note

Finance planning · margin on standard_unit_cost

40.2%

Gross profit $5.12m ($0.39m / 8.1% YoY)
margin -0.9 pp YoY · pack-adjusted

AP / Ops · margin on landed_unit_cost

37.4%

Landed gross profit $4.78m ($0.03m / 0.6% YoY)
2.8 pp below the Finance view; shown as a sensitivity

Sales ops · latest ingested_at wins

499,665 events

9,815 colliding IDs: newest version kept, 9,815 older moved to triage
Net sales $12.79m ($1.23m / 10.7% YoY)

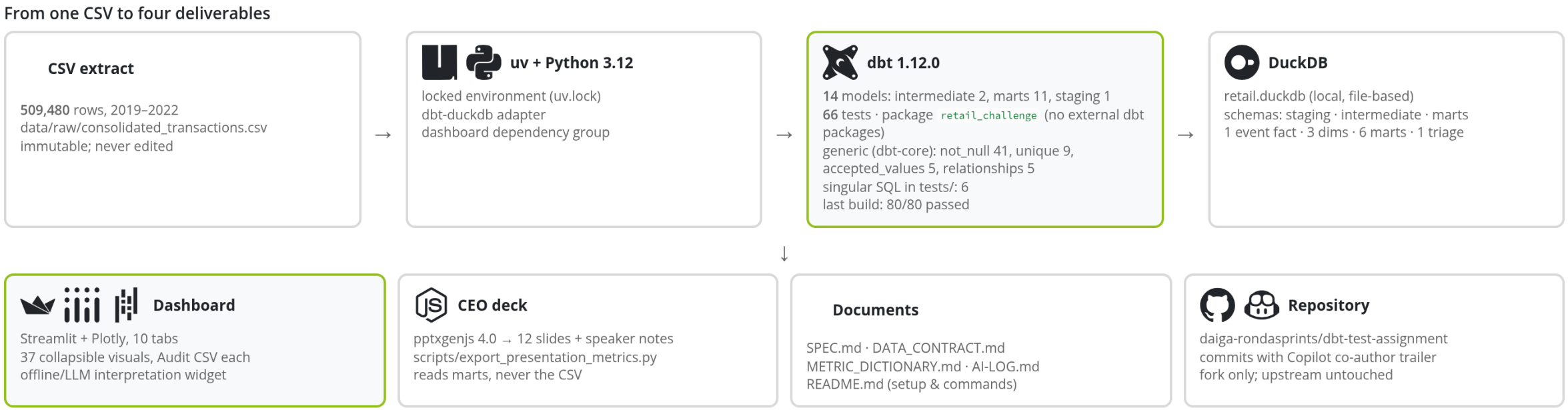
FP&A vs Store Finance · opex read as ANNUAL

$1.76m

Gross profit less store opex $3.35m; 121/287 stores cover it
Monthly reading rejected: x12 = $40.1m, only 4 stores would cover it

01 Solution map

From one CSV to four deliverables



Vendor marks: Simple Icons (CC0); trademarks belong to their owners. Counts are read live from dbt's manifest.json and run_results.json.

What do the 66 dbt tests check?

60 are dbt-core generic tests declared in YAML (not_null, unique, accepted_values, relationships) and 6 are singular SQL assertions: staging row count = source, unique row fingerprint, fact = selected versions, one triage row per collision, scorecard grain, and key agreement. No third-party dbt packages are needed yet (dbt_utils / dbt_expectations are on the roadmap).

66 tests: mostly not-null and uniqueness guards, plus 6 business assertions

Count by test type, from target/manifest.json



Deliverables list

Deliverable	Where	What it answers	How to regenerate
dbt project	models/, tests/, dbt_project.yml	Staging → versions → fact + dims → marts + triage	uv run dbt build
DuckDB database	retail.duckdb	All built relations; the single source for deck and dashboard	created by dbt build
Dashboard	dashboard/app.py	This app: model, row flow, team views, analysis, outliers	uv run --group dashboard streamlit
CEO deck	presentation/retail_ceo_debrief.pptx	12 slides with speaker notes	uv run python scripts/export_preser
Specification	SPEC.md	Grain, decisions, evidence, known limits	edit
Data contract	DATA_CONTRACT.md	What upstream must deliver in the transactions file	edit
Metric dictionary	METRIC_DICTIONARY.md	Global formulae and team views	edit
AI log	AI-LOG.md	How AI assistance was used and validated	edit

02 Data & model

What is in the file?

One CSV, 509,480 rows, 2019–2022: one sales line per row with product and store attributes repeated on every row.

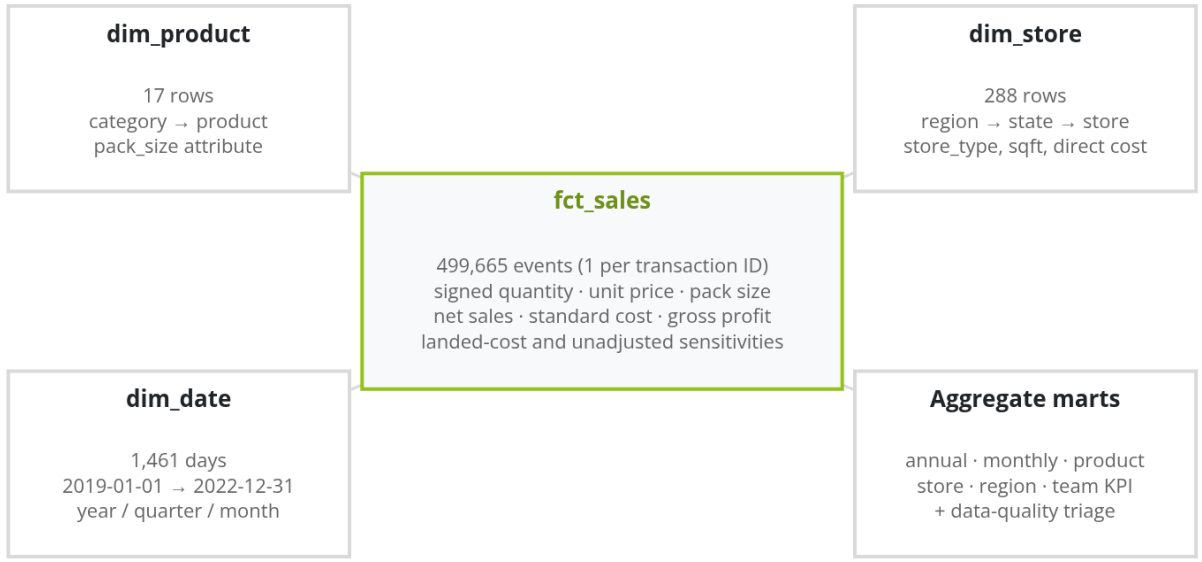
Column in the file	What we understand	How the model uses it
Transaction ID	Sales-line identifier. Not unique: 9,815 IDs appear twice.	Event key, after keeping the newest ingested_at version
Transaction Date	Economic date of the sale. Only 48 distinct dates, all the 1st of a month.	date_key; YoY and seasonality
quantity	Signed units; negative = return/adjustment (4,907 lines). Eaches vs packs un	Net sales, volume; kept signed
unit_price	Selling price per quantity unit.	Net sales = quantity x unit_price; average selling price
unit_cost	Third cost column; equals standard_unit_cost on ~98% of rows.	Not used; contract asks upstream to drop or define it
standard_unit_cost	Finance's budget cost. About 2x landed on pack_size 2, so read as a per-pacl	Primary gross profit: quantity x standard / pack_size
landed_unit_cost	What AP actually paid per unit.	AP/Ops margin sensitivity
pack_size	Units per pack (1 or 2); one value per product.	Divides standard cost; provisional pack equivalents
Product index	Product ID. Null or 99901 = unknown (1,410 rows).	product_key (-1 for Unknown)
Product Name	Name on the transaction; can disagree with the catalog.	Conflict detection only
product_name_catalog	Catalog name. Product 19 carries two names (Vanities / Cosmetics).	dim_product name (modal profile)
product_category	Six categories.	Top of the product hierarchy
product_effective_from / to	Intended validity window; identical for both product-19 names, so not a real	Not used for SCD2 (evidence below)
Store ID	Store key. Null or 99901 = unknown (1,408 rows).	store_key (-1 for Unknown)
store_name	Mostly a city name; not unique (273 names for 287 stores).	Label; city proxy in seasonality view
state_code / state	US state.	Store hierarchy: region > state > store
region_name	Census region; disagrees across rows for one Maryland store.	Recomputed from state_code
store_type	Nine civic types (City, Township, Borough, CDP, ...).	Slicer, not a hierarchy level
store_sqft	Floor area; one store reports 0.	Sales, cost and gross profit per sqft
direct_store_operating_costs	One figure per store, period unstated. Read as annual (see headline tile).	Store contribution scenario
source_system	POS / ERP / Manual lineage.	Tie-break when two versions share ingested_at
ingested_at	Load timestamp (lineage, not economic time).	Newest version wins on colliding IDs

How many facts and dimensions? Event or snapshot? Kimball or OBT? Type 2?

1 event fact (499,665 rows) and **3 dimensions**: product 16+Unknown, store 287+Unknown, date 1,461 days. Six marts on top are aggregates, not more facts.

A Kimball star: one event fact, three conformed dimensions

Marts are derived from the fact, never from the CSV



Event fact or snapshot fact?

Event. Rows are sales/returns, not balances. All 48 dates are the 1st of a month, so it looks monthly, but the monthly marts are aggregates of events, not a periodic snapshot.

Kimball or One Big Table?

Kimball. Product and store attributes conflict inside the file; conformed dimensions resolve each entity once, an OBT would sum the conflicts.

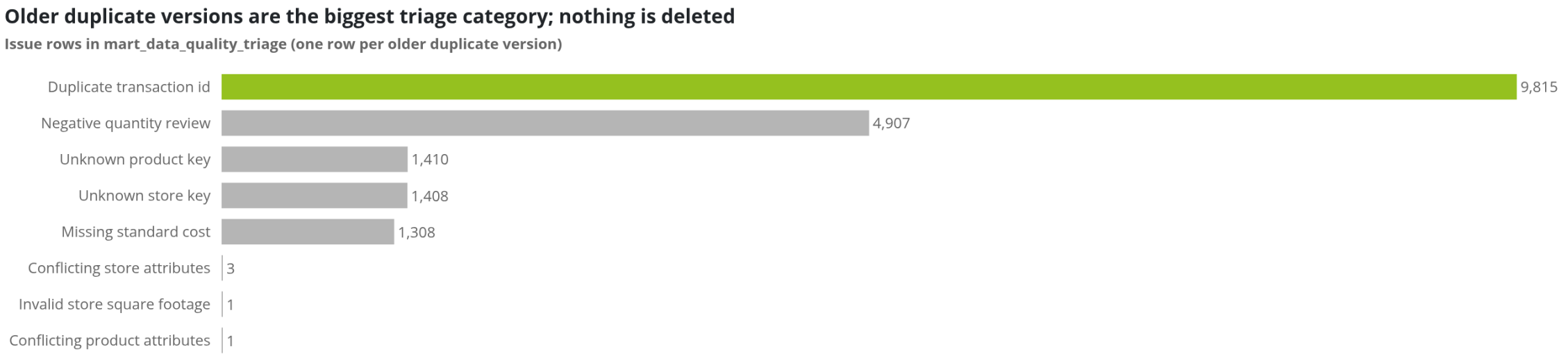
Type 2 dimension?

Not yet. The effective dates are not a history: product 19 is 'Vanities' and 'Cosmetics' in every year with the same window. Type 1 now; SCD2 when upstream sends real valid_from/valid_to.

product_name_catalog	product_effective_from	product_effective_to	first_seen
Cosmetics	2020-06-01 00:00:00	2022-12-31 00:00:00	2019-01-01
Vanities	2020-06-01 00:00:00	2022-12-31 00:00:00	2019-01-01

What happens to dirty data?

Nothing is deleted. Colliding IDs: newest `ingested_at` version goes to the fact, the older one to `mart_data_quality_triage`. Unknown keys, returns and missing costs stay in the fact and are flagged (18,853 issue rows; rebuilt every run). `DATA_CONTRACT.md` states what we expect upstream.



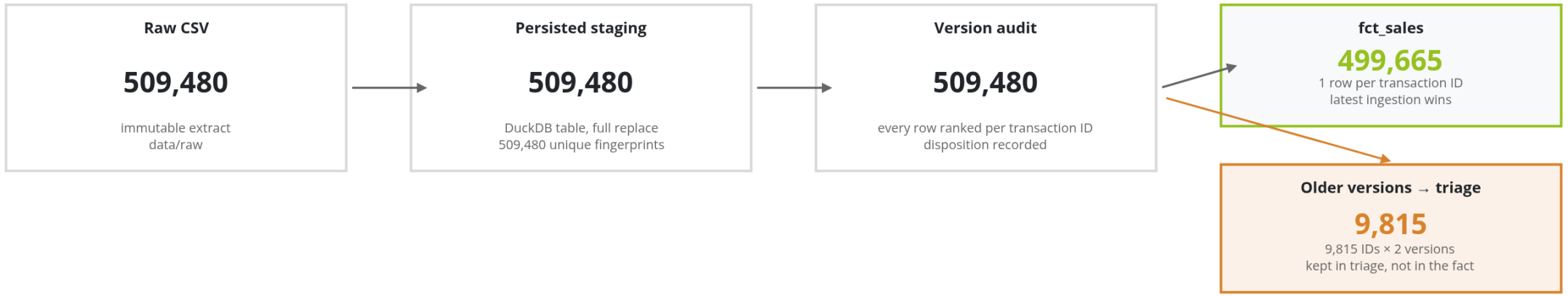
03 Row flow

How many rows did we start with, and how many were de-duplicated or sent to triage?

Start **509,480** → staging **509,480** (1:1) → version audit **509,480** → fact **499,665**. The only rows leaving are the **9,815** older versions of the 9,815 IDs that appeared twice. Triage flags 18,853 issue rows, but those are flags on retained rows.

Only older duplicate versions leave the pipeline; every other row reaches the fact

Row counts per layer. Triage also flags 18,853 issue rows on retained rows (unknown keys, returns, missing cost).



Is staging persisted and rerun-safe? Dimensions, facts, aggregates, date range, date dimension, derived metrics

Persisted staging, rerun-safe?

Yes. `stg_consolidated_transactions` is a DuckDB **table** rebuilt by full replacement on every `dbt build` ; two builds both landed 509,480 rows. Tests: staging count = source, fingerprint unique, fact = selected versions. Re-running cannot create duplicates (full reload, not CDC).

Dimensions and rows

Three: product 17 (16+Unknown), store 288 (287+Unknown), date 1,461 (2019-01-01 to 2022-12-31).

Fact tables and aggregates

One atomic fact, 499,665 rows. Six aggregate marts (annual, monthly, product, store, region, team KPI) so every team reads the same formulae; the fact stays the source of truth.

Must every query carry a date range?

No, a whole-fact sum is valid. A business report always should: it fixes the YoY period, excludes partial periods and lets the engine prune on `date_key` . The marts bake the calendar year in.

Date dimension?

Yes, `dim_date` : 1,461 days with year, quarter, month and weekday flags; the fact joins on an integer key.

Derived metrics

Net sales; standard cost of sales (pack-adjusted); gross profit and margin; YoY in $ and %; average selling price and a price/volume bridge; landed and unadjusted sensitivities; sales, cost and gross profit per sqft; pack equivalents (provisional); return-line rate; cost coverage; store-opex contribution. Formulae in `METRIC_DICTIONARY.md` .

team_view	metric_code	metric_value	metric_unit	metric_basis	formula
CEO	gross_margin_standard	0.4016	ratio	global	standard gross profit / standard-cost-covered net sales
CEO	gross_profit_standard	5124558.45	USD	global	net sales - standard cost of sales on cost-covered rows; standard cost adjuste
CEO	net_sales	12792067.24	USD	global	sum(signed quantity * unit_price)
CEO	net_sales_yoy_change	1231668.52	USD	global	this calendar year net sales - prior calendar year net sales
CEO	net_sales_yoy_pct	0.1065	ratio	global	net sales YoY change / prior calendar year net sales
Finance	gross_margin_landed_sensitivity	0.3735	ratio	landed_cost_sensitivity	landed gross profit / net sales
Finance	gross_margin_standard	0.4016	ratio	global	standard gross profit / standard-cost-covered net sales
Finance	gross_margin_standard_unadjusted_sensitivity	0.3014	ratio	pack_size_sensitivity	unadjusted standard gross profit / standard-cost-covered net sales

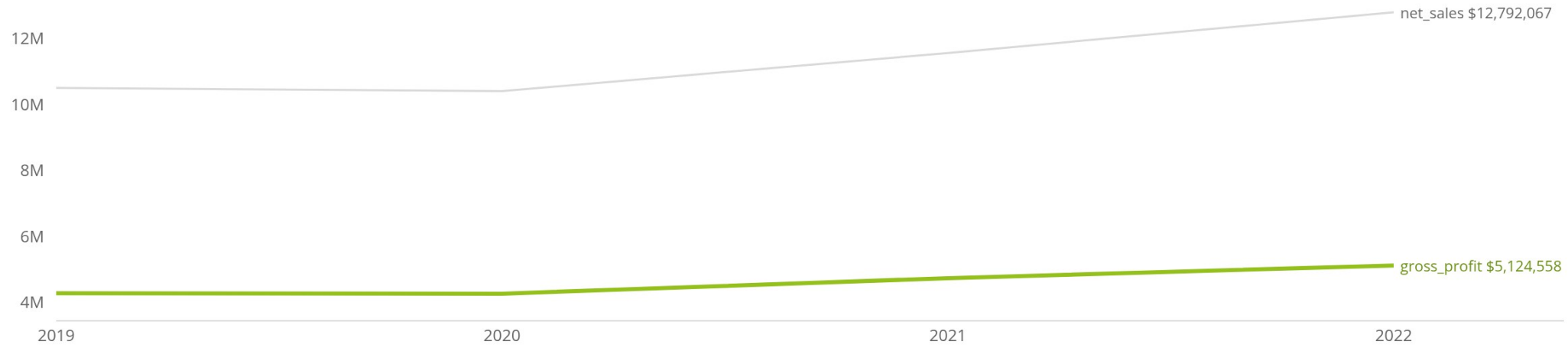
04 Management

Are we growing profitably?

Net sales $12.79m ($1.23m / 10.7% YoY); gross profit $5.12m ($0.39m / 8.1% YoY); margin 40.2%, -0.9 pp YoY. Growth is real but margin is slipping: profit grows slower than sales.

Sales 10.7% and gross profit 8.1% YoY in 2022; margin -0.9 pp

Company-wide net sales and standard gross profit by calendar year



Where did the growth come from: price or volume?

Of the $1.23m increase, $1.11m (90%) is volume and $0.12m is price/mix. ASP moved $32.18 → $32.49. The price/mix residual blends list-price moves with category mix; it is not price elasticity.



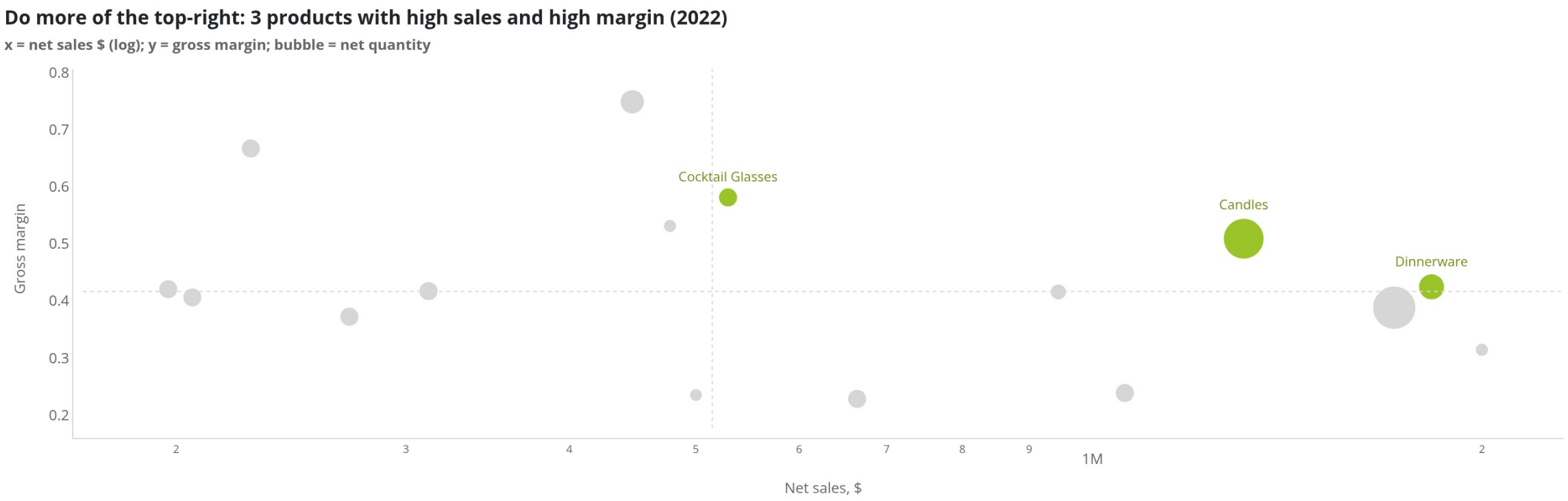
Which regions carry the company?

West delivers 72% of gross profit at 48.3% margin; Northeast is smallest at 24.5%. Concentration risk and the margin gap are both management questions.



What is working, and what should we do more of?

Top-right: above-median sales *and* margin — **Cocktail Glasses, Dinnerware, Candles**: more space, stock and promotion, in the store types and regions where their margin is highest. Bottom-right (high sales, low margin: Audio, Computers, Bathroom Furniture, Dining Furniture, Cookware) earns revenue but needs cost or price work before scaling.



After store operating cost, what is left?

Reading store opex as annual (decision in SPEC.md): $1.76m remains after $3.35m of direct store cost. South is the weakest region on contribution. Not net profit: central costs and tax are absent.



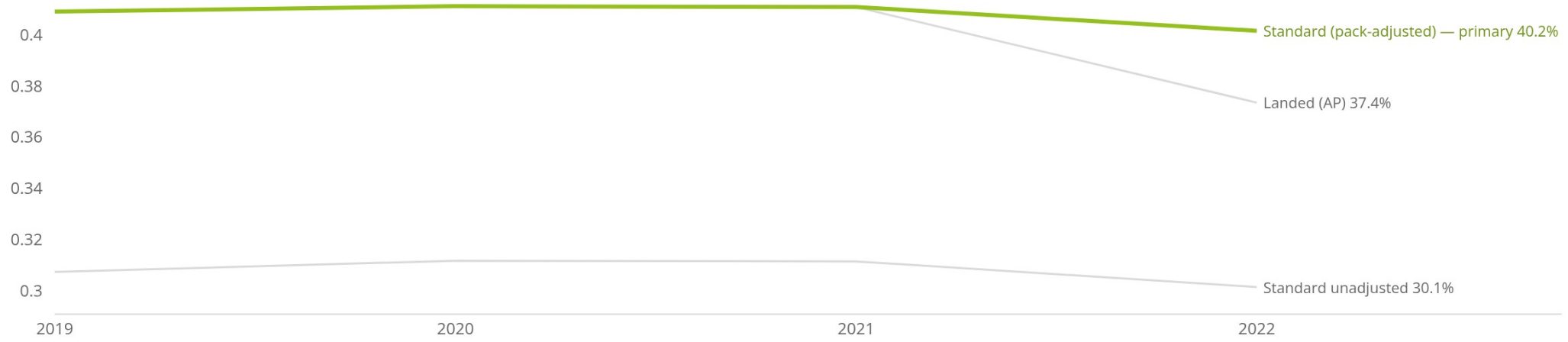
05 Finance

Which cost basis, and does the answer change with it?

Finance's **standard_unit_cost** is the primary basis (budget). Divided by pack_size it gives 40.2%; left unadjusted it gives 30.1%; AP's landed cost gives 37.4%. The level differs; the 2022 direction does not.

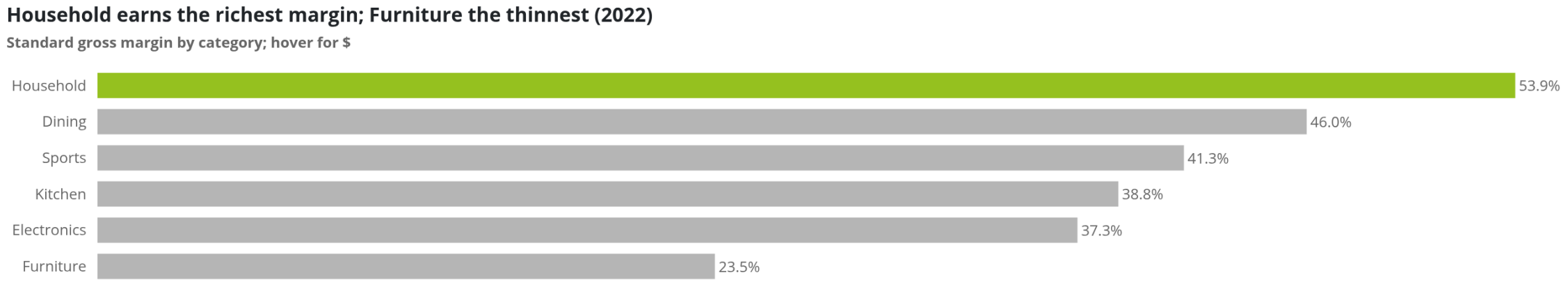
Three cost bases, one direction: margin fell in 2022 on all of them

Gross margin by calendar year and cost basis



Which categories make the margin?

Household leads at 53.9%; Furniture trails at 23.5% while contributing $2.23m of sales. Mix shift toward the low-margin categories explains part of the margin slide.



How much of sales has no standard cost?

99.74% of 2022 events carry a standard cost; $0.03m of sales is uncosted and excluded from the margin denominator (never from sales). The data contract asks for a cost on every line.

Uncosted sales by year

Net sales on rows with no standard_unit_cost



Store opex: annual or monthly? We say annual.

Read as annual, direct store cost is $3.35m (26% of sales) and 121/287 stores cover it from gross profit. Read as monthly it would be $40.1m — 8x gross profit, with only 4 stores covering it. Annual is the only reading consistent with a going concern; monthly is kept as a labelled sensitivity.

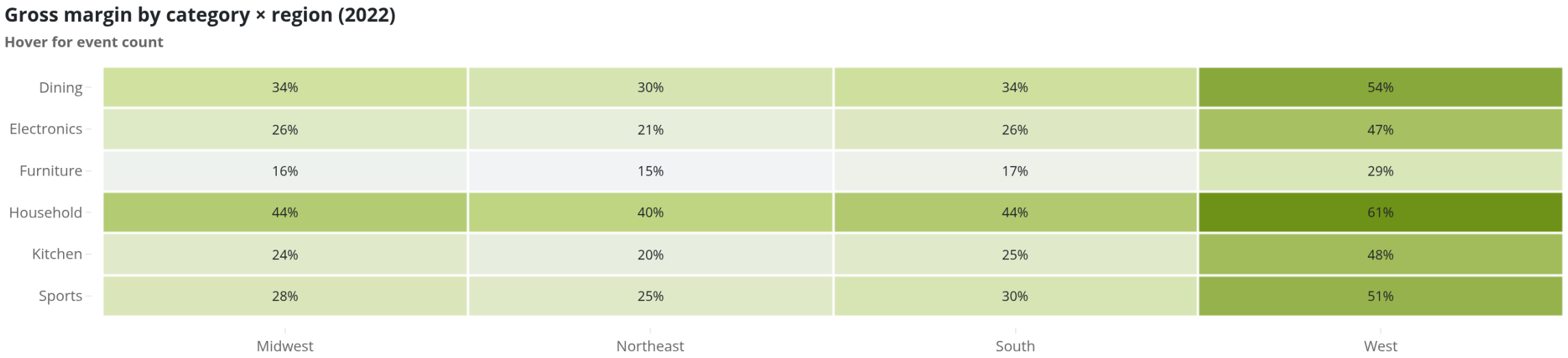
The monthly reading would dwarf gross profit; annual is the credible one

2022, stores with sales



Where is margin made: category × region

Same shared formula in every cell; the darkest column is the region to learn from.



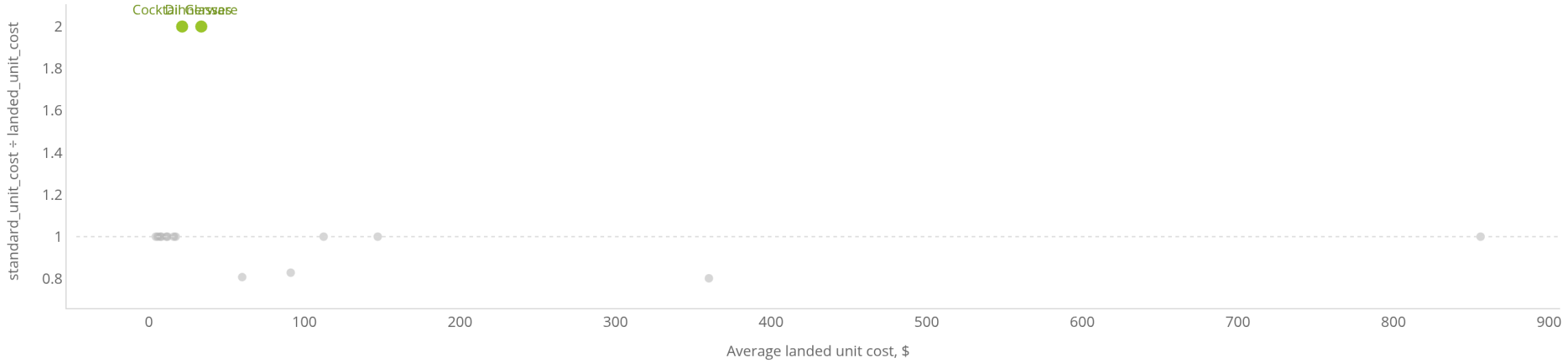
06 AP / Ops

How does what we paid (landed) compare with the budget cost (standard)?

For pack-size-1 products standard ≈ landed (median ratio 1.00). For pack-size-2 products the ratio is 2.00: standard cost is quoted per pack, landed per unit. That is why the model divides standard cost by pack_size before computing margin.

Pack-size-2 products carry a standard cost about twice their landed cost

x = average landed unit cost $; y = standard ÷ landed



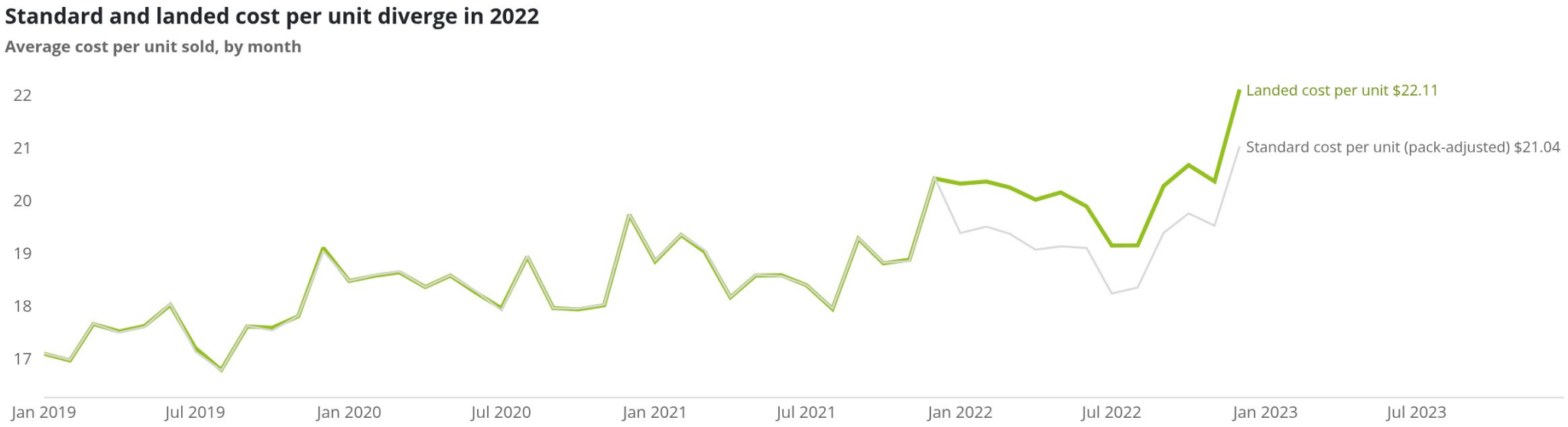
Margin on what we actually paid, by category

On landed cost, Household earns 53.9% and Furniture 7.4%. Compare with the Finance tab: the ranking is the same, the level is lower.



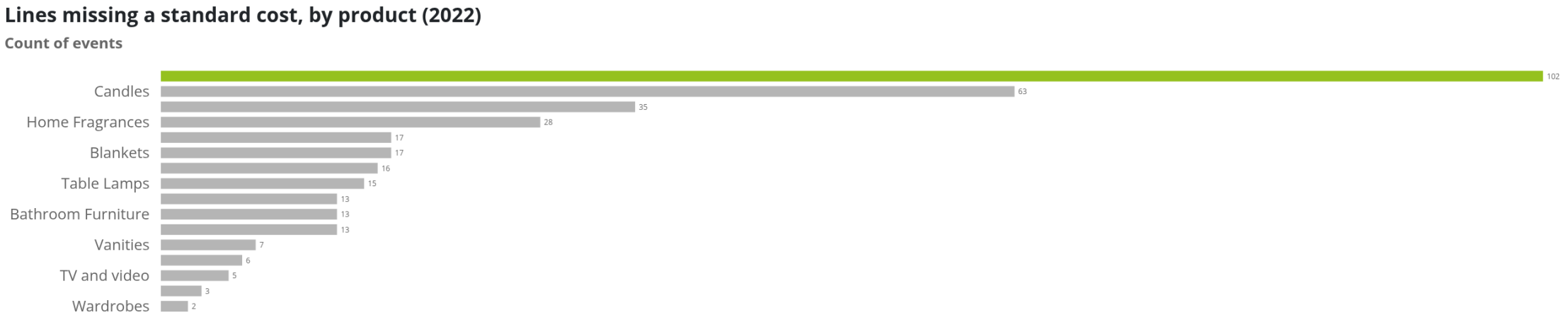
Did landed cost move away from standard in 2022?

Average cost per unit on the two bases tracked closely until 2022; by Dec 2022 standard is -4.9% vs landed. That divergence is the margin gap AP sees and Finance does not.



Which lines have no standard cost at all?

355 2022 lines ($0.03m of sales) have no standard_unit_cost; they still carry a landed cost. These are the rows AP can cost for Finance.



Where are returns and adjustments concentrated?

1.0% of 2022 lines are negative-quantity; Electronics has the highest rate. The source gives no reason code or link to the original sale — asked for in the data contract.

Return / adjustment line rate by category (2022)

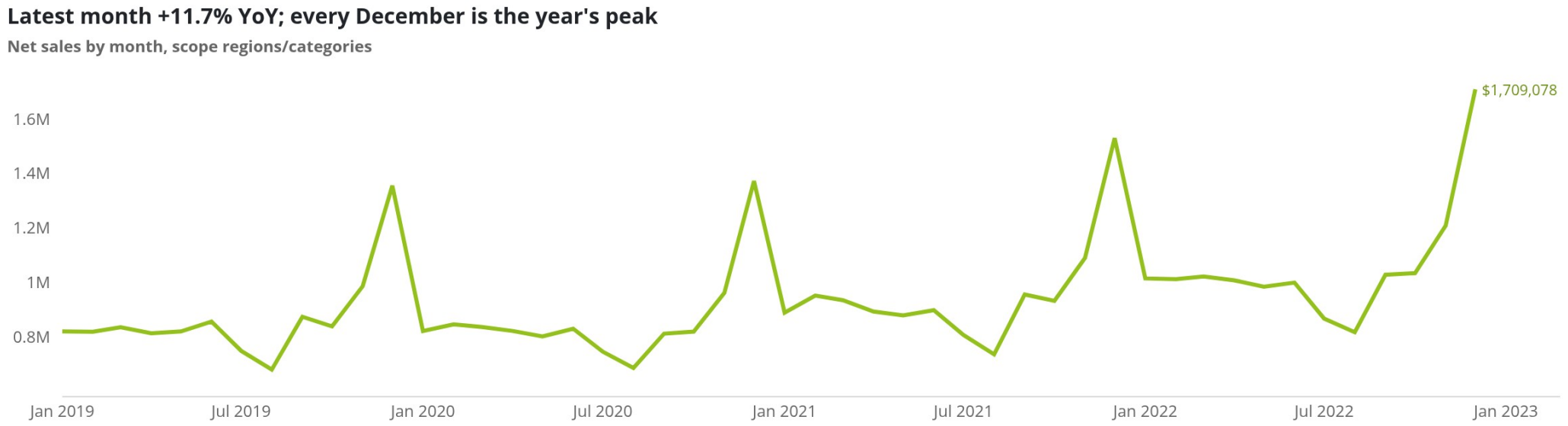
negative-quantity lines ÷ all lines



07 Sales

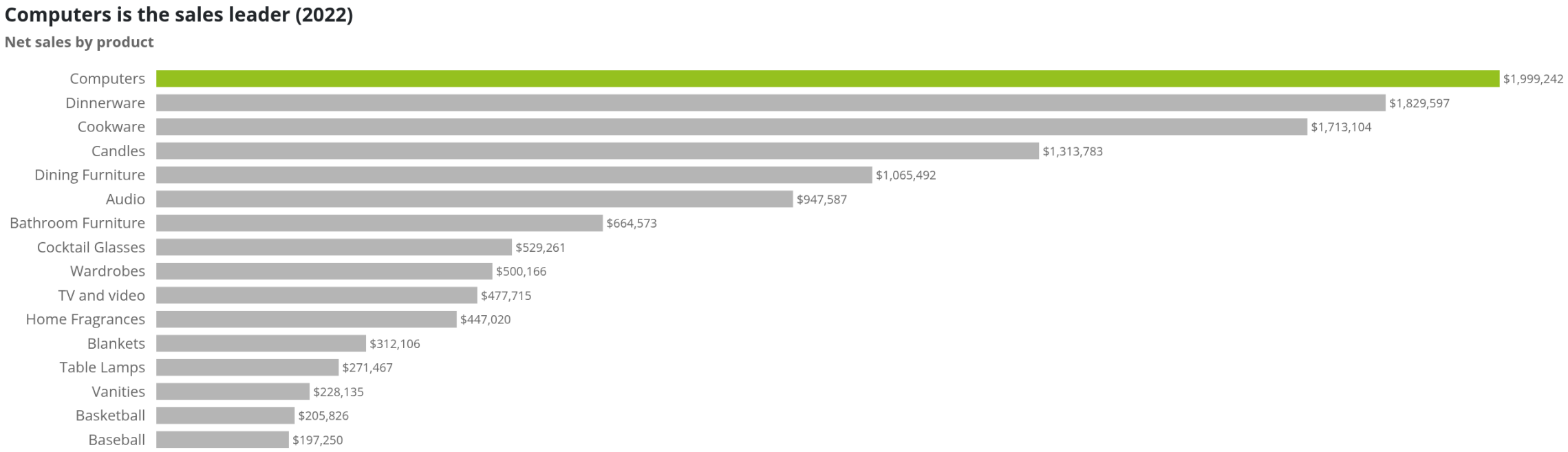
How are monthly sales trending, and what is the YoY?

Dec 2022: $1.71m, +11.7% vs the same month a year earlier. December is the peak every year; the trough is August. The audit CSV carries YoY for every month.



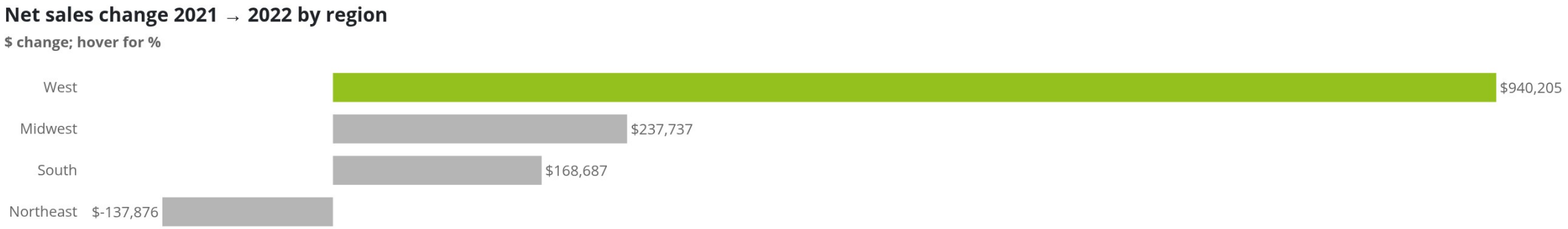
Which products sell the most?

Computers leads with $2.00m (16% of sales). The top three are Computers, Dinnerware, Cookware.



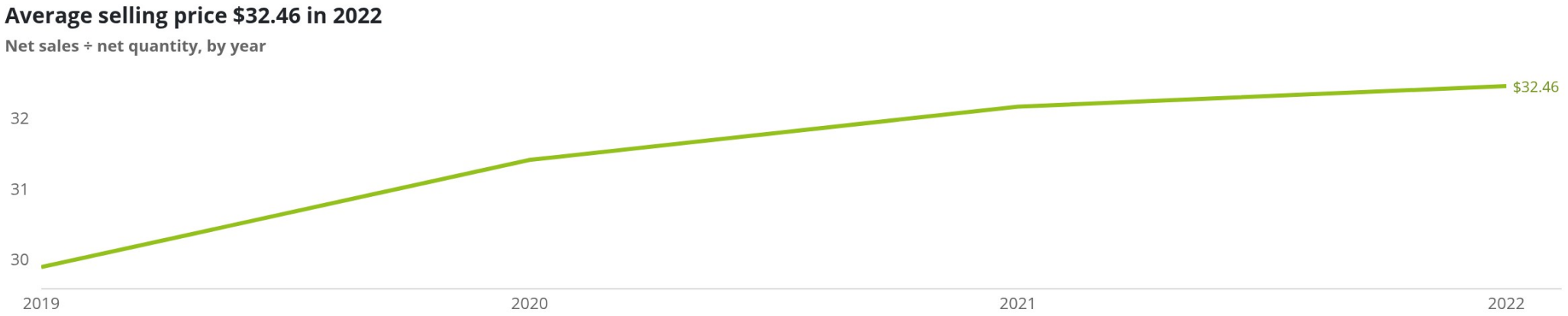
Which regions grew, and which shrank?

West added $0.94m; Northeast lost $0.14m (-8.8%).



Talk about price: what is happening to the average selling price?

ASP moved from $29.89 to $32.46 across 2019–2022; the latest step is +0.9%. Derived price metrics: price/mix effect (Management tab), sales per event, sales per pack-equivalent (provisional).



Sales-ops hygiene: sales per event, returns, collisions

Net sales per event

$94.77

134,982 events in 2022

Return / adjustment line rate

1.0%

1,357 negative-quantity lines

Colliding transaction IDs

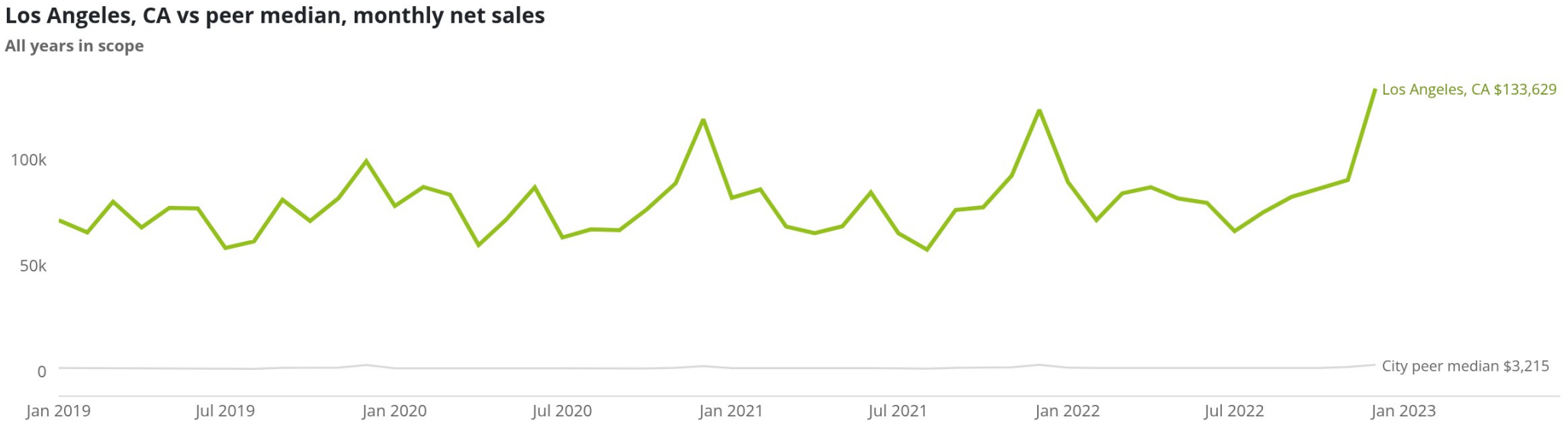
9,815

newest ingested_at kept; older to triage

08 Stores

How does this store track against its peers month by month?

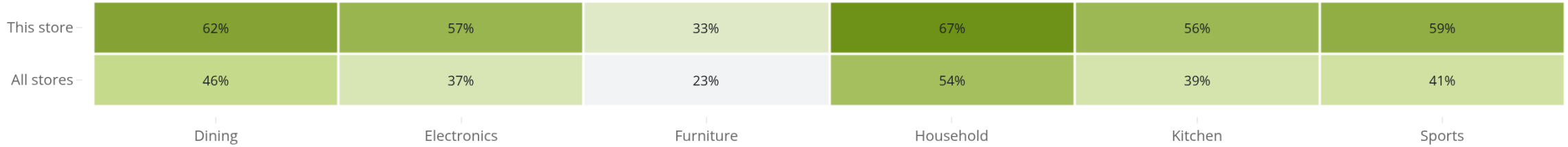
The accent line is Los Angeles, CA; grey is the median City store. A persistent gap is structure (size, location); a one-off spike is an event worth asking the store about.



Which categories does this store do better or worse than the company?

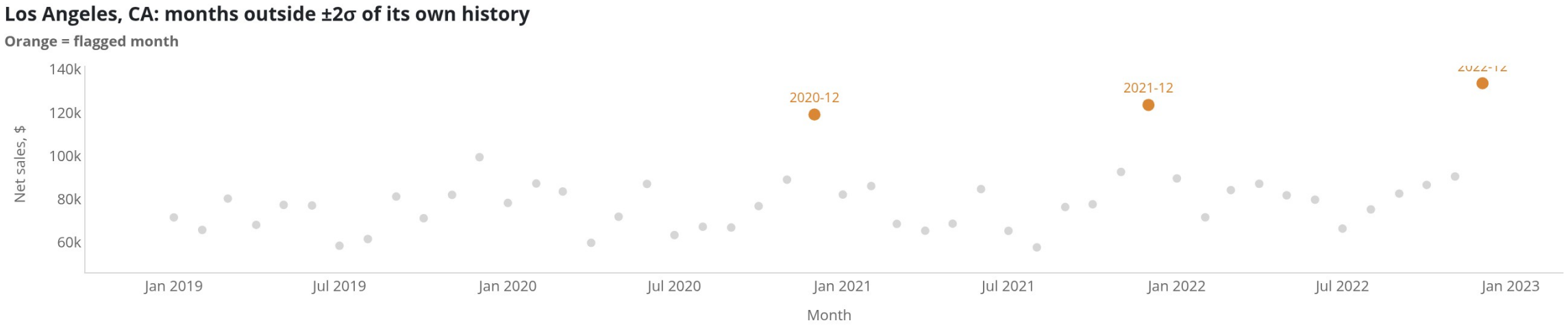
Row 1 is this store, row 2 the company. Cells darker in row 1 than row 2 are the store's strengths; the store sells 6 of 6 categories.

Gross margin by category: this store vs all stores (2022)



Which months broke this store's own pattern?

3 month(s) sit more than two standard deviations from the store's own mean: 2020-12, 2021-12, 2022-12. Decembers are expected; anything else deserves a question.



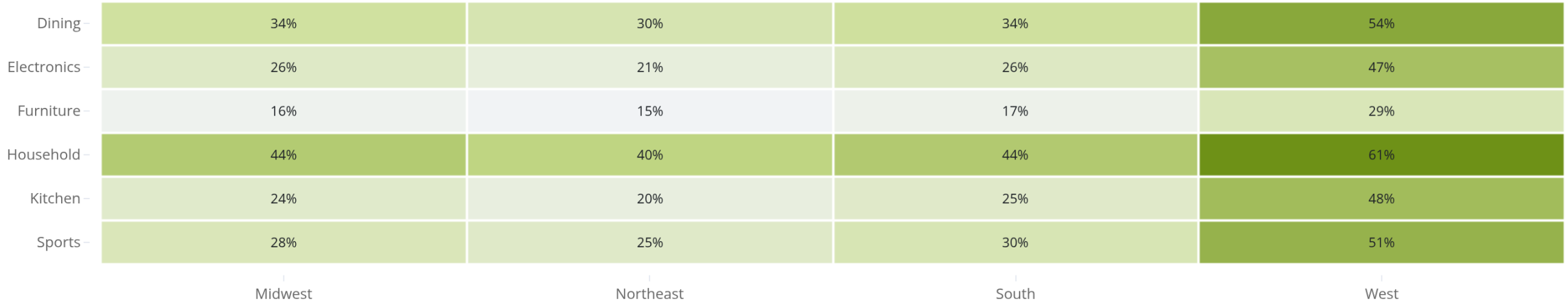
09 Analysis

Which product category is profitable where?

In 2022 every category earned positive gross profit in every region. **West** has the highest margin in 6 of 6 categories, and **CA** leads gross-profit dollars in 6 of 6. **Furniture** is the thin category everywhere.

West out-earns the other regions in 6 of 6 categories (2022)

Gross margin by category × region; hover for event count



Analysis | dashboard capture | gross profit, not net profit

Are larger stores performing better than smaller stores?

quartile	stores	avg sqft	avg net sales / store	net sales / sqft	gross margin
Q1 smallest	72	83,563	$16,424	$0.20	40.68%
Q2	71	139,328	$28,709	$0.21	42.86%
Q3	71	174,477	$38,347	$0.22	42.17%
Q4 largest	72	265,770	$94,480	$0.36	38.45%

They sell more, not more profitably. Area correlates with net sales (r = 0.54); the largest quartile averages $94,480 per store vs $16,424, with higher sales per sqft ($0.36 vs $0.20). But margin is **38.5%** vs **40.7%**. Size buys volume, not margin.

Bigger stores sell more per square foot, but margin does not follow (2022)

Each point is a store; accent = top 3 sales per sqft; hover for margin



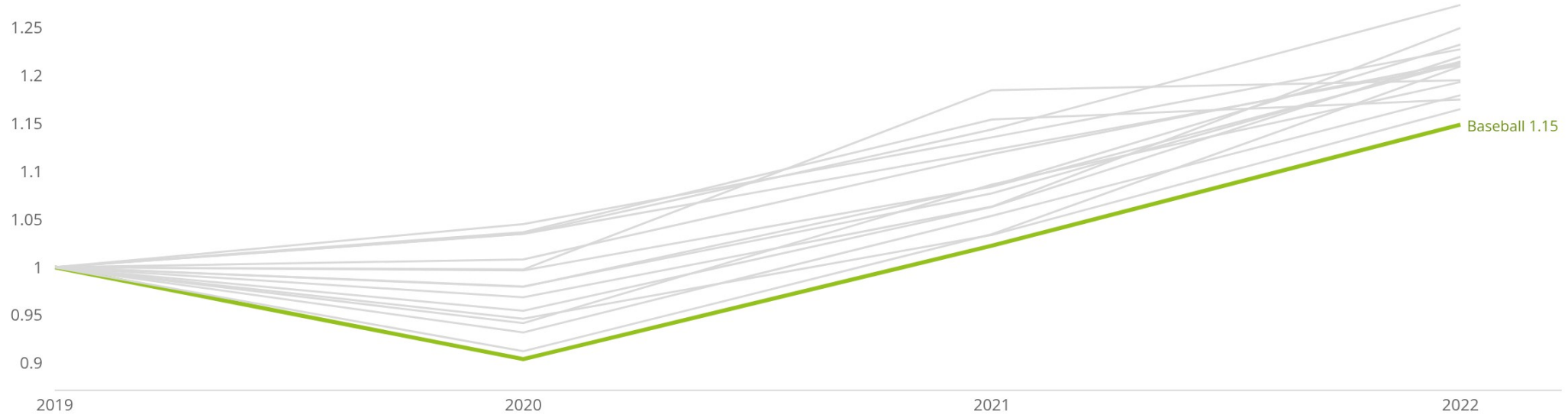
Dotted lines are medians. Louisville (store 200) has 0 sqft and is excluded from area ratios, not from sales.

Has any product fallen out of favour so that we lose sales consistently?

No. No product declined in every year 2019–2022, and 16 of 16 grew in 2022. The slowest is **Baseball** at 115% of its 2019 sales. The 2020 dip was broad and recovered everywhere.

Every product is above its 2019 level; Baseball grew least

Net sales indexed to 2019 = 1.00



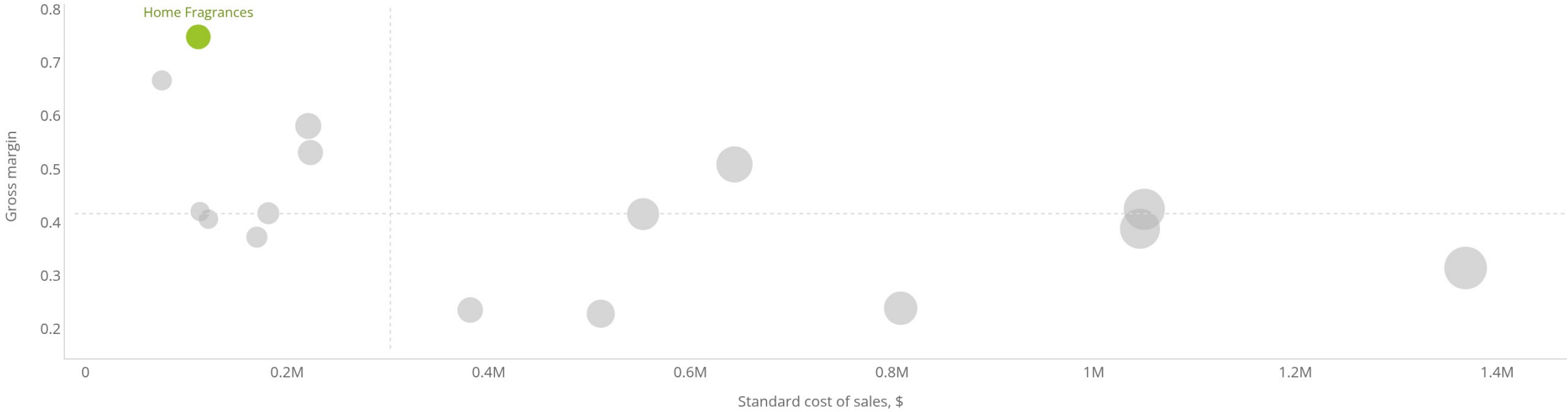
Analysis | dashboard capture | gross profit, not net profit

Where has our cost been lowest but margin highest?

Cost and margin are one ratio (margin = 1 − cost ÷ sales), so read cost *per sales dollar*. **Home Fragrances** spends 25.1% of each sales dollar on standard cost and earns 74.9% on $0.45m in 2022. Absolute lowest cost belongs to the small-volume lines — a different question.

Home Fragrances converts the least cost per sales dollar into the highest margin (2022)

x = standard cost of sales $; y = gross margin; bubble = net sales



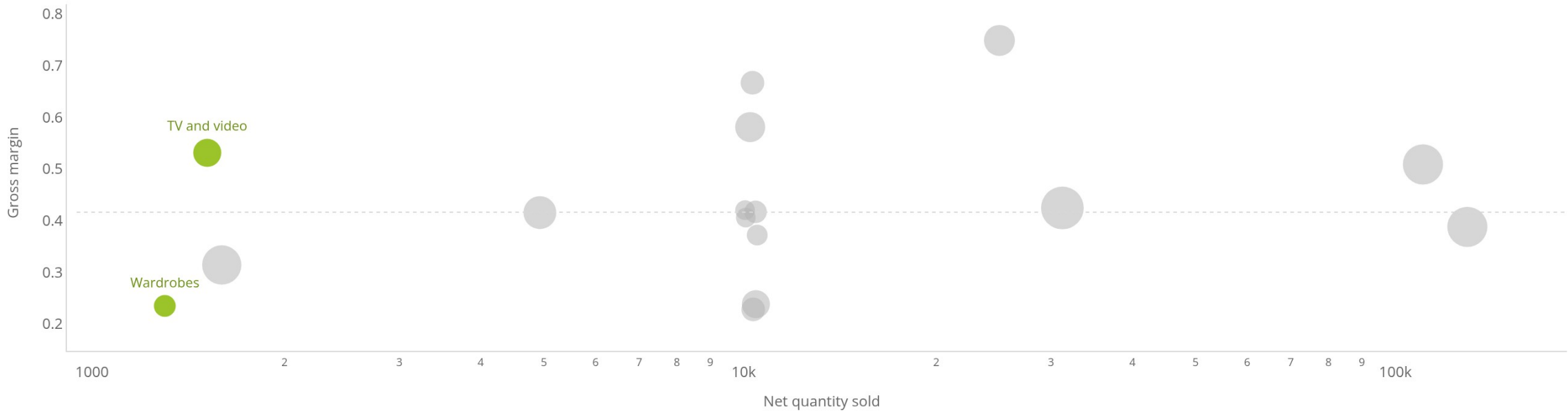
Analysis | dashboard capture | gross profit, not net profit

Which product sold the lowest quantity but gave us the highest margin?

Lowest quantity in 2022: **Wardrobes** (1,310 units, 23.5%) — low volume but *not* high margin. In the low-volume quartile, **TV and video** pairs few units (1,522) with 53.1% margin ($0.25m gross profit): the 'small but rich' line to protect.

TV and video: few units, high margin (2022)

x = net quantity (log); y = gross margin; bubble = gross profit $



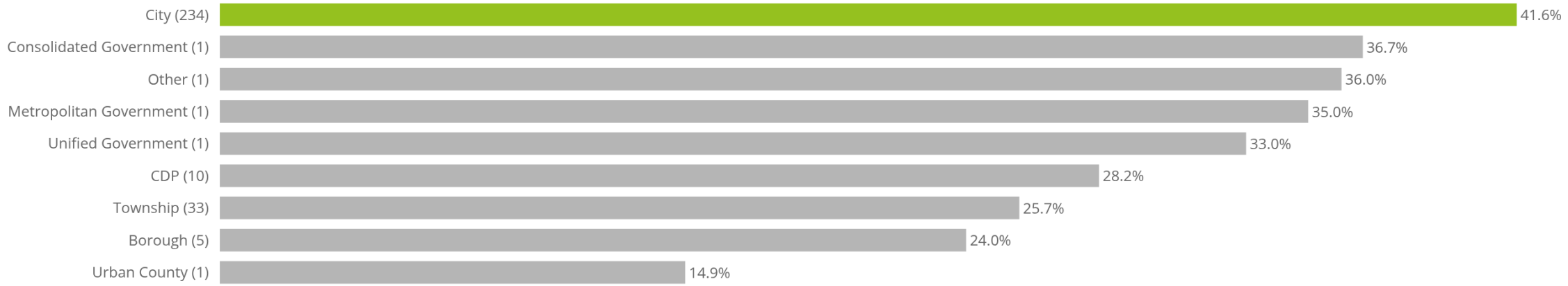
Analysis | dashboard capture | gross profit, not net profit

Is there a correlation between store type and profitability in general?

A pattern, not a proven correlation. City stores (234) earn 41.6%; the other multi-store types (Borough, CDP, Township) sit 13–18 points lower. 5 of 9 types have one or two stores, so their margins are single-store facts. Store type is a slicer to investigate, not a cause.

City stores lead on margin; most other types are too small to judge (2022)

Gross margin by store type (store count in brackets)

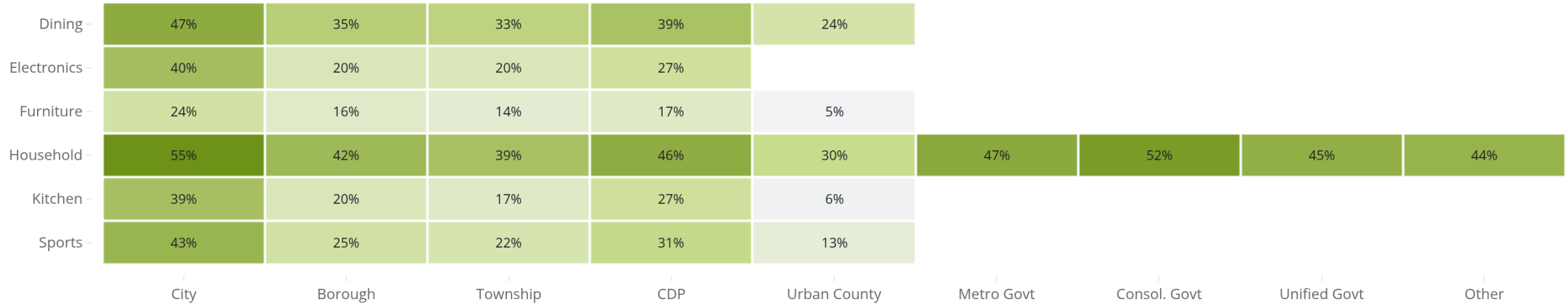


Is there a correlation between store type and profitability for each product category?

City stores have the best margin in **6 of 6** categories; the premium is widest for **Kitchen** (33 points) and narrowest for **Furniture** (19). **Furniture** is thin in every store type. Cells under 30 events are hidden.

City stores lead margin in 6 of 6 categories; the gap is largest in Kitchen (2022)

Gross margin by category × store type (≥ 30 events), columns by volume



Is there a correlation between time of year and certain products' sales?

Yes, strongest for big-ticket lines. Most products peak in **Dec** and trough in **Aug**. The swing is largest for **Computers** (1.91× its average in the best month) and smallest for **Cookware**. Four cycles and first-of-month dating: a strong pattern, not yet a forecast.

December lifts everything; Computers swings the most

Monthly net sales ÷ product's monthly average, all years in scope

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Computers	0.89	1.01	1.00	0.90	0.88	0.94	0.76	0.73	0.92	0.98	1.08	1.91
TV and video	0.93	1.03	0.92	0.87	0.88	0.89	0.82	0.68	0.95	0.94	1.22	1.86
Wardrobes	0.89	0.89	0.91	0.95	0.87	0.96	0.78	0.78	0.91	0.93	1.17	1.95
Bathroom Furniture	0.98	0.98	0.94	0.95	0.99	0.96	0.86	0.72	1.03	1.00	1.11	1.49
Blankets	0.92	0.95	0.95	0.97	0.91	0.95	0.87	0.80	1.00	0.98	1.13	1.57
Table Lamps	0.96	1.01	0.94	0.96	0.94	0.97	0.82	0.75	1.04	0.97	1.13	1.51
Cocktail Glasses	0.97	0.88	0.96	0.95	0.96	0.97	0.91	0.78	1.01	0.91	1.15	1.54
Home Fragrances	0.99	0.96	0.98	0.95	0.91	0.95	0.89	0.78	1.00	0.95	1.12	1.53
Dining Furniture	0.96	0.94	1.00	0.96	0.92	0.97	0.87	0.76	1.02	0.94	1.16	1.50
Baseball	0.98	0.99	0.99	0.95	0.92	1.00	0.80	0.78	0.94	0.98	1.17	1.52
Basketball	0.98	1.03	0.94	0.93	0.88	1.04	0.81	0.78	0.98	0.95	1.16	1.52
Audio	0.93	0.97	0.99	0.92	0.99	0.96	0.85	0.79	0.98	1.00	1.10	1.52
Vanities	1.00	0.96	0.99	0.94	0.95	0.93	0.80	0.83	0.96	0.95	1.17	1.53
Dinnerware	0.97	0.94	0.96	0.96	0.93	0.96	0.87	0.83	1.00	0.96	1.14	1.49
Candles	0.96	0.98	0.97	0.95	0.96	0.96	0.88	0.80	0.99	0.97	1.12	1.46
Cookware	0.94	0.96	0.94	0.98	0.95	0.95	0.89	0.82	0.98	0.99	1.15	1.44

Is there a correlation between time of year and a product's sales in certain states or cities?

Product

Computers

Level

State Store (city-name proxy)

States are a clean hierarchy level; the file has no city field, so *store name* is the closest proxy. If the December lift is similar in every row, the seasonality is national, not local.

Computers: seasonality by state (top 12 by sales)

Monthly net sales ÷ that row's monthly average; same scale as above

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
CA	0.93	0.99	1.07	0.88	0.78	0.96	0.69	0.73	0.74	1.01	1.03	2.21
NY	0.79	1.12	0.93	1.01	0.84	1.13	0.85	0.77	1.00	1.01	1.15	1.39
NV	0.90	1.01	1.02	0.96	0.94	0.86	0.75	1.00	1.03	0.89	1.15	1.48
IL	0.79	0.92	1.29	0.84	1.36	0.76	0.59	0.70	0.72	0.87	1.30	1.86
AZ	0.89	1.19	0.72	0.82	1.03	0.86	1.09	0.61	1.14	0.83	0.85	1.96
CO	0.87	1.14	0.84	0.71	1.16	1.00	0.97	0.57	0.97	1.19	0.74	1.83
VA	0.71	0.99	0.91	0.71	0.94	0.87	0.61	0.95	1.06	1.03	1.52	1.69
NC	1.04	0.46	1.12	0.85	0.83	0.72	1.04	0.39	1.16	0.83	1.42	2.13
HI	0.80	0.73	1.22	1.03	0.89	1.06	1.28	0.57	1.14	1.14	0.90	1.22
WA	0.80	1.08	0.91	0.61	0.86	1.07	0.81	0.44	0.61	1.35	1.24	2.23
LA	0.90	1.32	0.88	1.00	0.77	1.08	0.61	0.81	1.38	0.79	1.11	1.35
MD	1.28	1.24	0.75	0.88	1.26	1.10	0.62	0.25	1.13	0.75	1.12	1.62

Scatter: $ by time for each store — which stores are outliers?

$ measure

net sales

Outlier rule

- (•) Against peers in the same month
- ( ) Against the store's own history

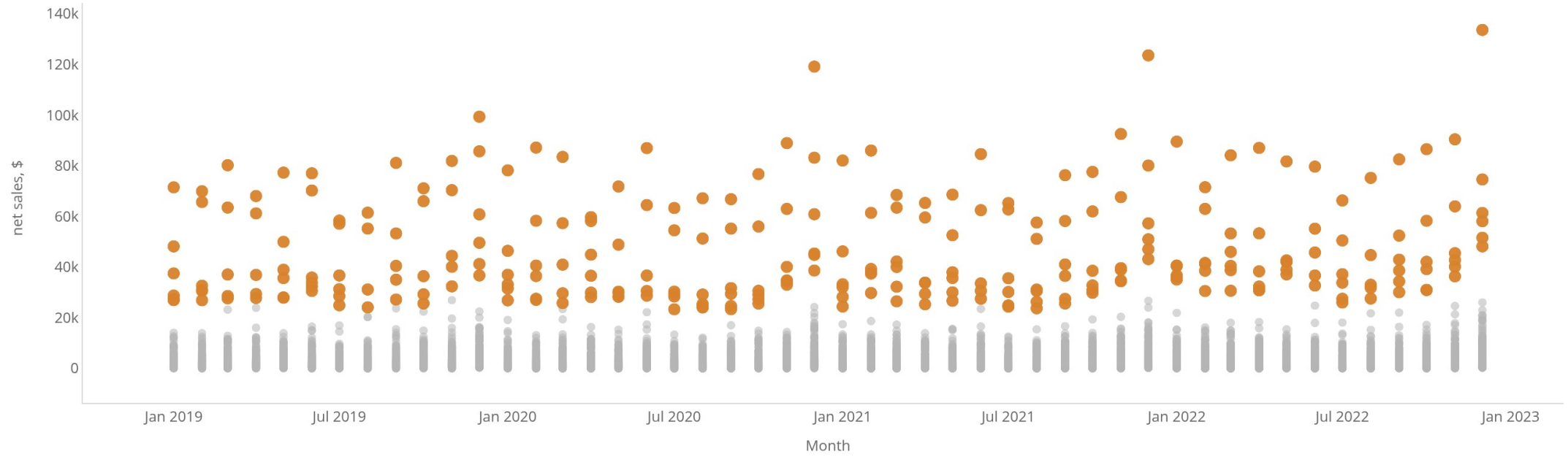
Flag when |z| exceeds

3.00

276 store-months exceed |z| 3.0 on net sales. Persistently **above** peers: Los Angeles, CA (48 months), New York City, NY (48 months), San Diego, CA (48 months), San Francisco, CA (46 months), Chicago, IL (43 months). Peer rule finds the structurally big winners; gross profit per sqft removes the size advantage.

Store-months outside |z| 3.0: 276 of 12,902

net sales per store per month; orange = flagged (against peers in the same month)

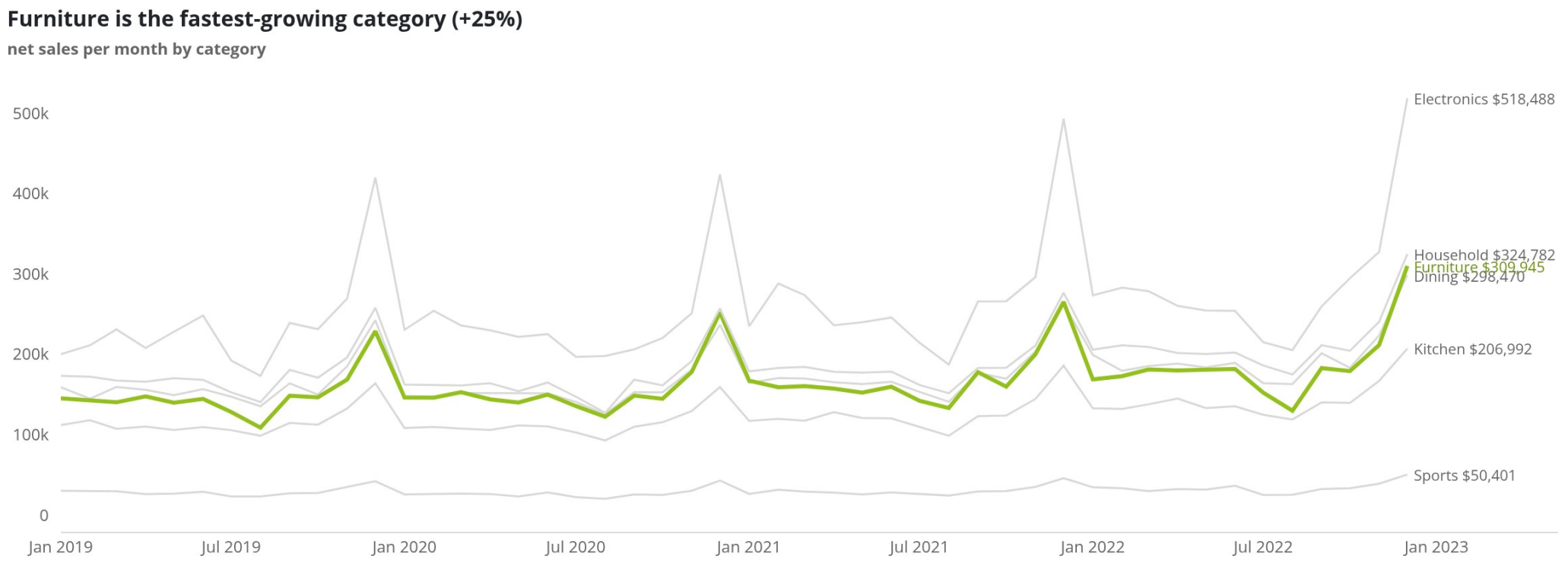


Scatter by product hierarchy: which categories or products are pulling ahead over time?

Product hierarchy level

(•) Category ( ) Product

Last 12 months in scope vs the first 12: **Furniture** grew most (+25%) and **Sports** least (+16%) on net sales. Every line shows the December spike; the gap between lines is the mix story.



Analysis | dashboard capture | gross profit, not net profit

10 Roadmap & AI

Future enhancements

Area	Enhancement	Why it matters	Effort	Depends on
Data contract	Line-level key and explicit unknown codes from upstream	Removes the fingerprint workaround and the 99901 guesswork	Low	Upstream team
Data contract	One labelled grain per cost column; drop unit_cost	Ends the pack-size inference for standard cost	Low	Finance / AP
Modelling	SCD Type 2 product and store dimensions	Historical attribute truth once valid_from/valid_to arrive	Medium	Upstream effective dates
Modelling	Incremental staging with merge on the line key	Daily loads without full reload; CDC-style lineage	Medium	Line key
Modelling	Periodic-snapshot fact for stock / store state	Answers inventory and days-of-cover questions	High	New source feeds
Metrics	dbt Semantic Layer / MetricFlow definitions	One governed formula set consumed by every BI tool	Medium	dbt Cloud or MetricFlow
Metrics	Confirmed store opex period and central cost allocation	Move from gross profit to a defensible contribution and operatin	Low	Store Finance / FP&A
Quality	dbt_utils and dbt_expectations tests; source freshness SLAs	Richer assertions (distributions, recency) without hand-written SC	Low	Package install
Quality	Profiling report per load (price/cost distributions)	Catch drift before marts rebuild	Medium	
Platform	Deploy to Snowflake / BigQuery with the same dbt project	Shared warehouse, scheduling, RBAC; DuckDB stays for local dev	Medium	Cloud account
Platform	CI with GitHub Actions: dbt build + dashboard AppTest on every P	Every change proves the pipeline and the app still work	Low	CI secrets
Dashboard	Row-level security per store and per team tab	Store managers see their store; finance sees all	Medium	Auth provider
Dashboard	Forecasting on the monthly series (seasonal naive / Prophet)	Turn the December pattern into a plan, with error bands	Medium	48+ months of data
Dashboard	PPT / Excel export per visual on the company template	Same chart in the deck and the app, one click	Low	Template file
AI	LLM interpretation on an approved endpoint, grounded in METRI	Natural-language answers that use the agreed formulae	Medium	Approved model, budget
AI	Anomaly narratives pushed to teams (weekly digest)	The outlier scatter becomes an alert, not a chart to remember to	Medium	Scheduler + email/Teams

Is there an AI inference widget? Yes — the Interpret button under every visual

Every chart has an **Interpret** button. Current engine: **offline interpreter (no data leaves this machine)**. The offline interpreter writes deterministic sentences from the chart's aggregate table (range, median, concentration, outliers, caveat) so the widget always works and no data leaves the machine. Set `AI_ENDPOINT`, `AI_API_KEY` and optionally `AI_MODEL` to route the same payload to an approved OpenAI-compatible endpoint (Azure OpenAI, GitHub Models). The payload is shown under each interpretation: question, action title, filters, caveat and at most 40 aggregated rows — never row-level data, SQL or credentials.

Demo interpretation

Across 6 rows, `net_sales` ranges from 403,076 to 3,424,543; the median is 2,294,544. The top item carries 27% of the total — spread out. Scope: year 2022. Figures are gross profit on standard cost unless the chart says otherwise; not net profit.

▸ {...}